

DATA AND BUSINESS ANALYTICS
DBA Project Report

Predictive Modelling on Referral Outcomes for Community Care Services

Prepared for
Agency for Integrated Care (AIC)

Submitted by
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Executive Summary

This report presents the findings of a predictive modelling project conducted in partnership with the Agency for Integrated Care (AIC), Singapore's national body overseeing community care services. Using anonymised referral data spanning January 2018 to December 2024 (580,000+ records), the team developed two complementary models to predict referral outcomes before resolution: a hierarchical ensemble pipeline for single-referral prediction, and a Deep Kalman Filter (DKF) temporal model for patients with repeated referrals. Exploratory analysis established the population context — 55% of referred patients are female, the Age 60+ group accounts for 91.6% of referrals (with the 75–89 band alone comprising nearly half), and only 53.7% of referrals result in care admission — framing the scale of the non-admission problem AIC faces.

The hierarchical ensemble (CatBoost + LightGBM at Level 1, CatBoost at Level 2) achieved a macro F1 of 0.401 and 66.9% overall accuracy. The Level 1 binary classifier alone achieved a macro F1 of 0.632, making it the most practically deployable output for AIC. Feature selection using Mutual Information, Random Forest Importance, and Cramér's V consistently identified service type (RS Svc Type Desc), provider frequency, and referral source frequency as the top predictors, accounting for over 30% of combined predictive signal. This indicates that referral outcomes are driven primarily by institutional and service-type factors rather than individual patient clinical profiles alone.

The DKF temporal model, trained on 72,751 repeat patients using 28 leak-free bio-psycho-social features, reached 69.26% accuracy against a 65.7% majority baseline. It identified three trajectory groups: Community Care Working (92–93% eventual acceptance), At Risk of Institutionalisation (92.4% eventual acceptance but with a declining trajectory), and Cycling Risk (67.8% eventual acceptance, with consistently low probability across all referrals). Trajectory divergence between groups becomes detectable at the 3rd–4th referral — the earliest actionable intervention window for caseworkers. Age at application is the strongest quantitative risk factor for cycling, and formal agency rejection accounts for only 0.3% of non-admission outcomes, confirming non-acceptance is overwhelmingly patient- or provider-initiated rather than the result of system failure.

Key limitations include the inability to predict rare outcomes (Reject By Agency is undetectable at 0.3% of cases), the dominance of provider-identity features over patient-level signal in the ensemble, and the absence of post-discharge longitudinal data. Recommendations for AIC are to deploy the Level 1 binary model as a practical early-warning referral screener, implement the DKF cycling-risk flag to trigger proactive caseworker review at a patient's 3rd referral, and invest in longitudinal data collection so future work can directly measure whether community care delays residential care placement.

1. Company Introduction

The Agency for Integrated Care (AIC) is a Singapore government body created by the Ministry of Health (MOH) that manages and coordinates community care services across the country. AIC acts as the central hub connecting patients, hospitals, and community care service providers — including day rehabilitation centres, nursing homes, home nursing services, and day care facilities.

When a patient requires community care, a referral is submitted through AIC's national system. AIC matches the patient with an appropriate provider based on clinical needs, living situation, and bio-psycho-social factors. The referral can result in admission, rejection, or withdrawal. Understanding which referrals are likely to fail is critical to AIC's mission: each unsuccessful referral represents a patient who may not receive timely care, potentially leading to delayed recovery or premature admission to residential care. With Singapore's rapidly ageing population, the ability to anticipate and prevent referral failures is of growing strategic importance.

2. Problem Definition and Motivation

2.1 Research Questions

This project addressed three core research questions posed by AIC: (1) What factors determine whether patients are accepted or rejected for community care? (2) Does access to community care delay the need for residential care (institutionalisation)? (3) What factors in patients who utilise community care delay residential care? The primary focus was Research Question 1, which is most directly addressable with the available data; Questions 2 and 3 are explored through the Deep Kalman Filter temporal model.

2.2 The Prediction Problem

Formally, the task is a multi-class classification problem: given patient demographic, clinical, and referral-level features available at the time of submission, predict which of seven outcome categories the referral will fall into. The dataset is severely imbalanced — Admit accounts for 71.6% of cases, Withdrawn for 22.8%, and the remaining five classes for under 6% combined, with Reject By Agency representing fewer than 60 cases in the entire dataset. The full outcome distribution is shown in Appendix A1, Figure A1-1, and class definitions are listed in Appendix A1.

2.3 Why This Problem Matters

A model predicting every case as “Admit” would achieve 71.6% raw accuracy while providing no actionable insight. The project therefore optimised for balanced accuracy and macro F1 — metrics that weight all classes equally regardless of size — ensuring the model performs meaningfully on non-admission outcomes, which are the cases most relevant for AIC’s intervention.

3. Methodology and Tools

3.1 Data Overview and Preparation

The dataset was assembled from four source tables provided by AIC — Caregiver Contact, Functional Status, Biodata, and Transaction RS (referral records). After merging, the dataset contained 580,000+ records and 46 features across 2018–2024. Two separate data pipelines were prepared: one for the hierarchical ensemble (all referrals; 294,201 rows after provider filtering), and one for the DKF (repeat patients with 3+ referrals; 72,751 patients, 206,034 referral rows). Detailed cleaning and transformation steps, including supporting figures, are provided in Appendix A2.

In summary, caregiver-contact rows where the patient was listed as their own caregiver were removed as data-entry errors. Caregiver relationships were one-hot encoded and a total caregiver count per patient was engineered. Functional status blanks — confirmed by AIC to represent non-applicable fields rather than missing data — were filled with “Not Applicable”. Patient age was calculated at referral submission date and grouped into 15-year bands; ages outside scope were excluded. From the referral records, sub-outcome fields were removed as they constitute post-resolution leakage, and referrals without an assigned service provider were excluded as they would skew the model toward non-admission outcomes. Four engineered features were added: service-provider frequency, referral-source frequency, binary “rare” indicators for each (threshold: <0.02% of all referrals), and a COVID-19 period indicator (January 2020–February 2023) to capture pandemic-related behavioural shifts.

3.2 Feature Selection

Feature selection was conducted on the training set only, using three complementary methods to prevent data leakage and ensure robustness: Mutual Information (MI), Random Forest Importance, and Cramér’s V. All three methods yielded highly consistent rankings. The top predictors were RS Svc Type Desc (service type requested), SP Full Name_freq (service-provider frequency), and Referral Source_freq — together accounting for over 30% of combined predictive signal across all methods. Clinical indicators (wound care, hearing/visual impairment, mobility, activity tolerance) and housing features (lift access, accommodation status) rounded out the top ten. Full method-by-method feature importance scores are provided in Appendix A3, Figure A3-1.

This finding has a key operational implication: referral outcomes at AIC are driven substantially by institutional factors — which service is requested and who provides it — rather than purely by individual patient clinical profiles. Provider-level patterns (which providers tend to accept which service types) encode much of the predictive signal.

For the DKF pipeline, a dedicated leakage-detection script was run across all 46 features, confirming two post-outcome columns as definitive leaks: RS SubOutcome Item Desc (100% of its values predict a single outcome class) and RS Last Status Datetime (96.7% of values predict a single class). Both were excluded. The DKF was trained on 28 confirmed safe bio-psycho-social features, supplemented by four engineered temporal features: submission year, submission month, days since first referral, and duplicate referral count per patient.

3.3 Modelling Approach

Two modelling approaches were developed in parallel. For the ensemble pipeline, two phases were conducted. In Phase 1, standard multi-class classifiers (XGBoost, Random Forest, Logistic Regression, LightGBM, CatBoost) were trained on all seven outcome classes with SMOTE oversampling; the best result was a macro F1 of 0.32 (Random Forest). Collapsing to three classes (Admit / Withdrawn / Other) improved macro F1 to 0.47, demonstrating that the data signal is clearer with simplified targets and motivating the hierarchical approach. In Phase 2, the problem was decomposed into a two-level hierarchy. Level 1 (binary routing) predicts Admit vs Non-Admit; Level 2 classifies Non-Admit cases into Withdrawn, Pre-assigned Withdrawn, or Other Non-Admit. Multiple pipeline versions (V3 through V7) were tested, experimenting with model combinations, threshold tuning, and class weighting. The

key finding was that Level 1 routing quality is the primary performance bottleneck — errors at this stage propagate to the final output. (Full Phase 1 model comparison is provided in Appendix A5.)

For the DKF, each repeat patient's referral history was treated as a time series. The model architecture comprises three neural network components: a GRU inference network that reads the full referral sequence and infers a latent state distribution at each timestep, a transition network (MLP with Tanh activation) that predicts how the latent state should evolve between referrals, and an emission network that maps the latent state to outcome class probabilities. The model was trained by minimising a combined loss of cross-entropy prediction loss and a KL divergence term that keeps the inferred trajectory consistent with the transition network's expectations. Latent dimensionality was selected across four configurations (8, 16, 32, 64) using the Akaike Information Criterion, penalising model complexity against fit improvement, with the winning dimension retrained for 30 full epochs.

3.4 Final Model Architecture

The ensemble pipeline uses a CatBoost + LightGBM ensemble at Level 1, averaging predicted Non-Admit probabilities with a tuned threshold of 0.55, and a standalone CatBoost model at Level 2. CatBoost was selected for its native handling of high-cardinality categoricals (SP Full Name with 380+ unique values, Referral Source) without requiring one-hot encoding; LightGBM complemented this with speed and strong binary calibration.

The DKF architecture uses a 2-layer GRU (hidden dimension of 64) as the inference backbone, producing per-timestep mean and log-variance parameters for the latent state via learned linear projections. The transition network is a 2-layer MLP

(Linear–Tanh–Linear) that predicts the prior latent state from the previous posterior, implementing the Kalman-style dynamics. The emission network (Linear–ReLU–Dropout(0.2)–Linear) decodes the latent state to 3-class outcome logits. Class imbalance (65.7% Accepted, 34.0% Withdrawn, 0.3% Rejected) was addressed through inverse-frequency class weights in the loss function (Rejected weight 50×, Withdrawn weight 2×). Sequences were padded to a maximum length of 25 referrals, with padding positions masked from the loss computation.

4. Main Results

4.1 Ensemble Model Progression

Table 1 tracks macro F1 improvement through successive model versions, demonstrating that hierarchical decomposition consistently outperformed flat classification. Although confidence-gating achieved higher accuracy, the ensemble model was selected for its more stable and consistent performance across classes.

Model Version	Accuracy	Macro F1	Key Change
Flat 7-class (best)	63.3%	0.318	SMOTE + Random Forest baseline
3-class simplification	54.2%	0.469	Collapse to Admit / Withdrawn / Other
V4 – 2-level hierarchy	54.4%	0.431	Hierarchical pipeline introduced
V5 – Confidence gating	72.6%	0.446	Probability calibration added
V7 – Tuned CatBoost	58.20%	0.407	Hyperparameter tuning
Final Ensemble	66.9%	0.401	CatBoost + LightGBM L1 ensemble

Table 1: Macro F1 progression across pipeline versions.

4.2 Final Ensemble Model Performance

The final ensemble hierarchical pipeline was evaluated on a completely held-out test set of 58,841 referrals. Results are shown in Table 2.

Metric	Level 1 (Binary)	Level 2 (Non-Admit)	Full Pipeline
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Accuracy	65.8%	71.3%	66.9%
Balanced Accuracy	66.7%	72.3%	47.0%
Macro F1	0.632	0.609	0.401
Weighted F1	0.674	0.747	0.667

Table 2: Final ensemble performance across the two-level hierarchy.

Level 1 achieved a macro F1 of 0.632 and balanced accuracy of 66.7% — a strong result for a binary classifier with a 72:28 class split. Level 2 achieved macro F1 of 0.609 and balanced accuracy of 72.3%, demonstrating that once restricted to non-admit cases, the model discriminates well between outcome types. The full-pipeline macro F1 of 0.401 is lower because routing errors at Level 1 compound into the final output: approximately 33% of non-admit cases are incorrectly routed to Admit at Level 1.

4.3 Key Predictors and SHAP Analysis

SHAP (SHapley Additive exPlanations) analysis was conducted on the Level 1 CatBoost model to interpret individual predictions. RS Svc Type Desc dominates across all outcome classes — certain service types strongly push predictions toward Non-Admit (e.g. Home Environment Review, Day Care), while others push toward Admit (e.g. Nursing Home, Meals-on-Wheels). SP Full Name frequency and Referral Source frequency are the next most influential features. Among patient-level predictors, accommodation status and age at application contribute most meaningfully. Full SHAP beeswarm and mean absolute SHAP plots, including service-type disaggregation, are provided in Appendix A3 (Figures A3-2 to A3-4). Together, these results reflect real-world differences in provider capacity and referral pathway maturity across service types.

4.4 Deep Kalman Filter — Temporal Predictive Model

Patient trajectory analysis classified all 72,751 repeat patients into three groups based on their acceptance probability trend across referrals. Community Care Working patients (stable or improving trajectory) achieved a 92–93% eventual acceptance rate, confirming that sustained community care engagement successfully delays institutionalisation for the majority. At Risk of Institutionalisation patients showed a declining trajectory but still reached 92.4% eventual acceptance with sufficient referrals. Cycling Risk patients — defined as those with 3+ referrals whose predicted acceptance probability never exceeded 50% — reached only 67.8% eventual acceptance, with 32.2% having no recorded admission across the 2018–2024 period.

Critically, trajectory divergence between groups becomes statistically distinguishable at the 3rd–4th referral, identifying this as the earliest actionable intervention window for caseworkers. Among quantitative risk factors, age at application is the strongest predictor of cycling risk (Cohen's $d = -0.25$, $p < 0.05$), with older patients at first application significantly more likely to enter a referral spiral — consistent with increasing clinical complexity in elderly patients. Service type is a parallel risk factor: Home Environment Review and Day Care referrals show disproportionately high non-admission rates, while Nursing Home and Meals-on-Wheels referrals show stronger admission rates, suggesting service-type matching is a key lever for reducing cycling risk.

5. Assumptions, Limitations, and Contributions

5.1 Key Assumptions

Sub-outcome fields were excluded as post-resolution data (data leakage). The COVID-19 period was defined as January 2020–February 2023. Referrals without an assigned provider at submission were excluded as structurally distinct cases.

5.2 Limitations

The most significant limitation is predicting rare outcomes: Reject By Agency (55 cases) and Withdrawn from SP Waiting List (454 cases) have insufficient data for reliable modelling. The dominance of provider-identity features over patient-level signal also means the ensemble model partly reflects provider behaviour rather than purely clinical risk. Finally, the absence of longitudinal follow-up data makes it impossible to directly answer Research Questions 2 and 3 — only proxies via repeat-referral patterns are available.

5.3 Addressing the Research Questions

Research Question 1 is directly answered: service type and provider characteristics are the dominant determinants of referral outcomes, with physical accessibility (lift access, accommodation) and clinical complexity (wound care, mobility) as the leading patient-level factors. Research Question 2 is addressed through the DKF temporal analysis: of 72,751 repeat patients modelled, 92–93% of non-cycling patients eventually achieve acceptance, while Cycling Risk patients reach only 67.8% — providing empirical evidence that community care engagement delays institutionalisation for the majority but fails for an identifiable subgroup. Research Question 3 is partially addressed: age at application is the strongest quantitative risk factor for cycling, and categorical analysis of service type confirms that Home

Environment Review and Day Care referrals are disproportionately associated with non-admission outcomes, suggesting service-type matching is a key lever AIC can use to reduce cycling risk.

5.4 Contributions

This project establishes a replicable, leak-free pipeline for AIC's referral data, demonstrating that hierarchical decomposition outperforms flat classification by ~25% on macro F1 and identifying service type and provider frequency as dominant predictors for intervention planning. The complementary DKF temporal model, achieving 69.26% accuracy on clean features, identifies a critical intervention window at the 3rd–4th referral for cycling-risk patients. Together, these tools provide AIC with both a static referral screener and a dynamic longitudinal risk monitor.

6. Conclusion

Working with 580,000+ records spanning seven years, this project delivered two complementary tools: a hierarchical ensemble model for single-referral prediction and a Deep Kalman Filter for tracking patients across repeated referrals. The Level 1 binary classifier (macro F1 0.632) is the most immediately deployable output and can serve as an early-warning screener. The DKF flags cycling-risk patients as early as their 3rd–4th referral, opening a concrete intervention window. The dominance of service-type and provider features over patient clinical signal, alongside the finding that formal agency rejection accounts for only 0.3% of non-admissions, indicates that AIC's most actionable lever is service-type matching and proactive caseworker engagement — not redesigning eligibility rules. Future work should focus on collecting longitudinal post-discharge data to directly test whether community care

engagement delays residential care placement, and on continued tuning to improve accuracy on the rare-outcome classes.

Appendix

A1. Outcome Class Definitions and Distribution

The seven referral outcome classes in the original dataset, prior to hierarchical consolidation, are listed in Table A1-1. Their relative frequency across the 2018–2024 dataset is shown in Figure A1-1 and motivates the severe class-imbalance challenge described in Section 2.2.

Outcome Class	Description	Proportion
Admit	Patient successfully admitted to the community care service	71.6%
Withdrawn	Referral withdrawn by patient or caregiver	22.8%
Pre-assign Withdrawn	Withdrawn after a service provider was tentatively assigned	2.0%
Pre Assign Withdrawn – BPR	Variant of pre-assignment withdrawal	1.4%
Withdrawn & Case Closed	Withdrawn and closed by the referring party	2.0%
Withdrawn from SP Waiting List	Withdrawn while on a provider's waiting list	0.15%
Reject By Agency	Service provider rejected the referral	0.02%

Table A1-1: Outcome class definitions and observed proportions.

Figure 1: Referral Outcome Distribution (2018-2024)

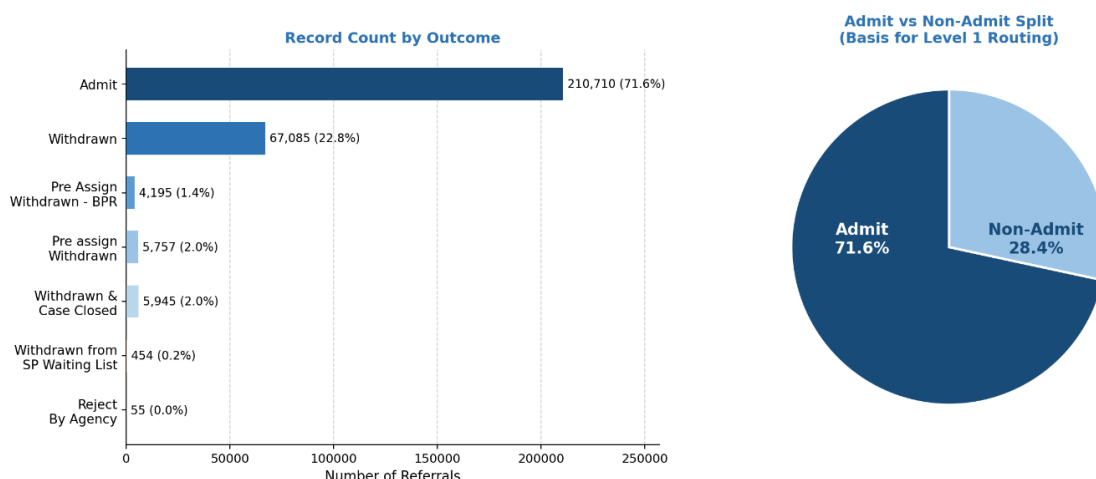


Figure A1-1: Distribution of referral outcomes across 294,201 records. The dominant Admit class (71.6%) and severe imbalance of non-admission outcomes present the core modelling challenge.

A2. Data Preparation Supporting Figures

The following figures illustrate key data cleaning and transformation steps described in Section 3.1.

Application ID	Care Giver Relationship
2018/0001	Daughter
2018/0002	Son
2018/0003	OtherRelative
2018/0004	Daughter
2018/0004	Daughter
2018/0005	Daughter
2018/0005	Son

Figure A2-1: Sample of raw caregiver relationship data before transformation.

Num of Caregiver	ct_brotherfamilyrelationship	ct_daughter	ct_daughter-in-law	ct_neighbour	ct_other relatives	ct_otherrelative	ct_parent	ct_sibling	ct_sisterfamilyrelationship	ct_son	ct_son-in-law
1	0	0	0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0	0	0
1	0	1	0	0	0	0	0	0	0	0	0
1	0	1	0	0	0	0	0	0	0	0	0
2	0	1	0	0	0	0	0	0	0	0	0
1	0	1	0	0	0	0	0	0	0	0	0
2	0	0	1	0	0	0	0	0	0	0	0
1	0	1	0	0	0	0	0	0	0	0	0
2	0	0	0	0	0	1	0	0	0	0	0
3	0	1	0	0	0	1	0	0	0	0	0
1	0	0	1	0	0	0	0	0	0	0	0
2	0	0	0	0	0	0	0	0	0	0	0
1	0	1	0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0	0	0
2	0	1	0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0	0	0
2	0	1	0	0	0	0	0	0	0	0	0

Figure A2-2: Caregiver data after one-hot encoding and merging by Application ID.

Application	AIC Log Aids Needs	AIC Log Respiratory Care	AIC Log Level of Assistance	AIC Log Feeding	AIC Log Mobility	AIC Log Toileting	AIC Log Transfer	AIC Log Hearing Impairment	AIC Log Mental Status	AIC Log Activity Tolerance
2018/0167	--	--	Minimal	Independent	Ambulant	Independent	Independent	No	Rational	Not Applicable
2018/0174	--	--	--	Dependent	Bed Bound	Dependent	Total Dependence	No	Unable To Respond	Not Applicable
2018/0646	--	--	--	Dependent	Bed Bound	Dependent	Not Applicable	Yes	Unable To Respond	Not Applicable
2018/0893	--	--	--	Dependent	Bed Bound	Dependent	Total Dependence	AIC Internal Use	Others	Not Applicable
2018/0988	--	--	--	Independent	Bed Bound	Dependent	Not Applicable	No	Rational	Not Applicable
2018/1004	--	--	--	Dependent	Bed Bound	Dependent	Total Dependence	No	Unable To Respond	Not Applicable
2018/10133	--	--	Moderate	Independent	Ambulant	Needs Assistance	Not Applicable	No	Rational	Not Applicable
2018/10193	--	--	Minimal	Independent	Ambulant	Independent	Not Applicable	No	Rational	Not Applicable
2018/10220	--	--	--	Dependent	Bed Bound	Dependent	Not Applicable	No	Rational	Not Applicable
2018/10335	--	--	--	Dependent	Bed Bound	Dependent	Total Dependence	No	Unable To Respond	Not Applicable

Figure A2-3: Functional status data before filling — blank cells visible.

AIC Log Respiratory Care	AIC Log Level of Assistance	AIC Log Feeding	AIC Log Mobility	AIC Log Toileting	AIC Log Transfer	AIC Log Hearing Impairment	AIC Log Mental Status	AIC Log Activity Tolerance	AIC Log Visual Impairment	AIC Log Wound Care
Not Applicable	Not Applicable	Independent	Bed Bound	Dependent	Moderate	No	Rational	Fair	No	No
Not Applicable	Minimal	Independent	Ambulant	Independent	Minimal	No	Rational	Good	No	No
Not Applicable	Maximum	Dependent	Wheelchair Bound	Dependent	Maximum	No	Rational	Fair	No	No
Not Applicable	Minimal	Independent	Ambulant	Needs Assistance	Minimal	No	Confused	Fair	No	No
Not Applicable	Minimal	Independent	Ambulant	Needs Assistance	Minimal	Yes	Rational	Good	No	No
Not Applicable	Minimal	Independent	Ambulant	Independent	Minimal	No	Others	Fair	No	No
Not Applicable	Moderate	Need Assistance	Ambulant	Dependent	Moderate	No	Confused	Poor	No	No
Not Applicable	Minimal	Need Assistance	Ambulant	Dependent	Minimal	Yes	Rational	Fair	Yes	No
Not Applicable	Not Applicable	Not Applicable	Not Applicable	Not Applicable	Not Applicable	Not Applicable	Not Applicable	Not Applicable	Not Applicable	Not Applicable
Not Applicable	Minimal	Independent	Ambulant	Needs Assistance	Minimal	No	Rational	Good	No	No
Not Applicable	Minimal	Independent	Ambulant	Dependent	Minimal	No	Rational	Fair	No	No
Not Applicable	Not Applicable	Dependent	Bed Bound	Dependent	Maximum	No	Others	Fair	No	No

Figure A2-4: Functional status data after replacing blanks with 'Not Applicable'.

A3. Feature Importance and SHAP Analysis

Figure A3-1 shows the full feature importance ranking referenced in Section 3.2;

Figures A3-2 to A3-4 show the SHAP analysis referenced in Section 4.3.

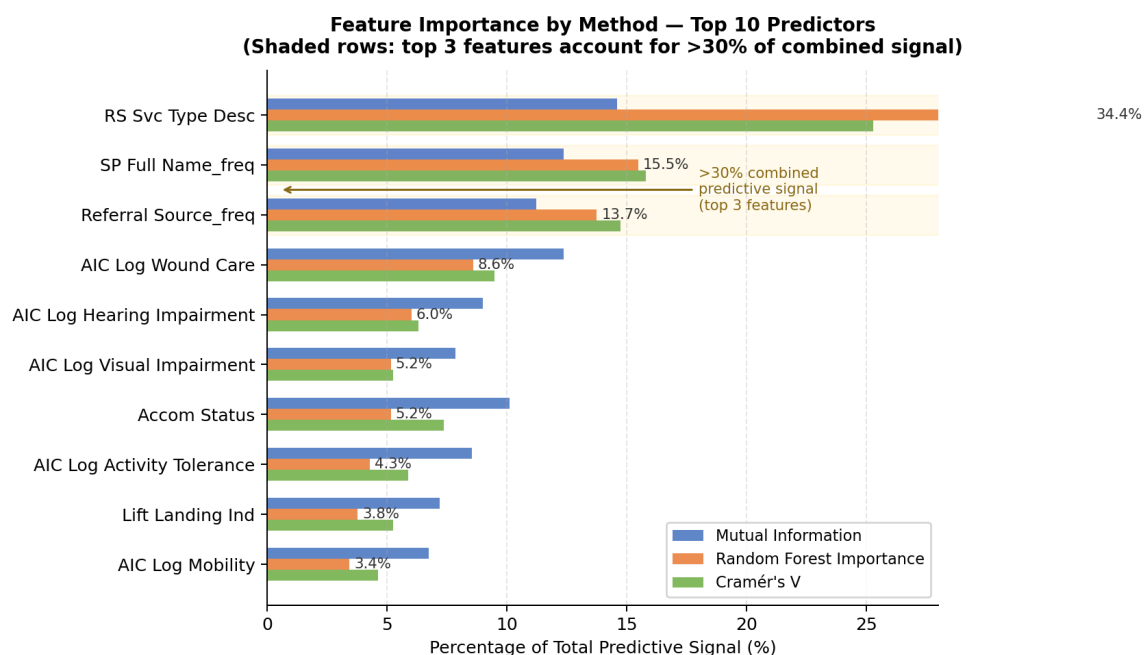


Figure A3-1: Feature importance scores across all three selection methods. The top three features — RS Svc Type Desc, SP Full Name_freq, and Referral Source_freq — account for over 30% of total combined predictive signal (highlighted rows).

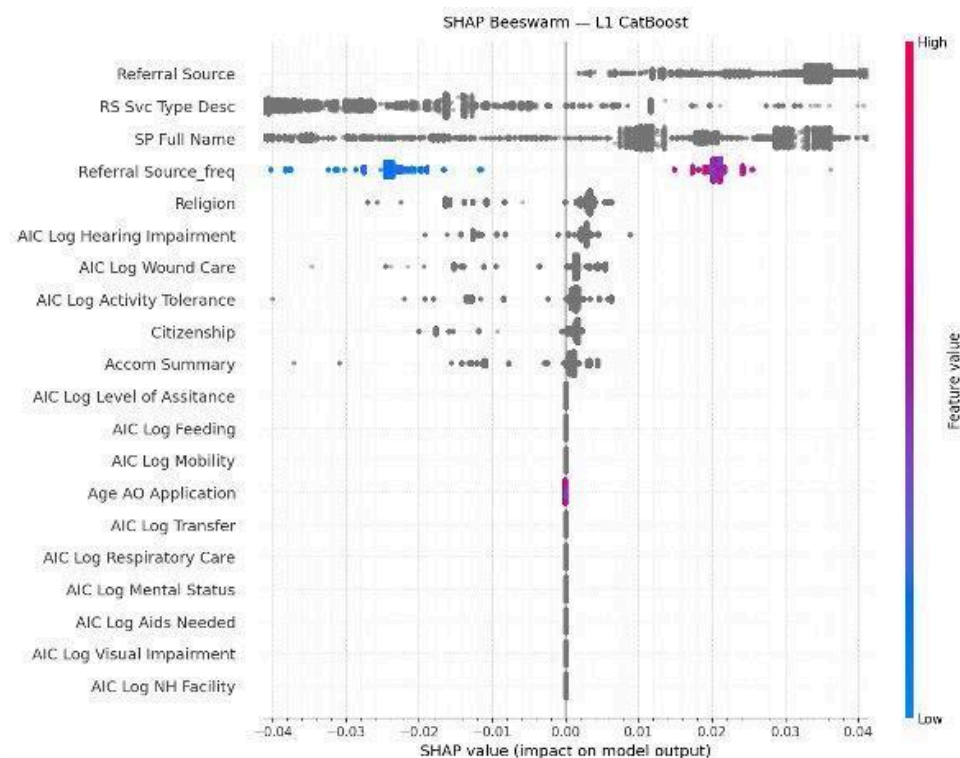


Figure A3-2: SHAP beeswarm plot for the L1 CatBoost model. Each point is one referral. Rightward position indicates higher Non-Admit probability; leftward indicates higher Admit probability.

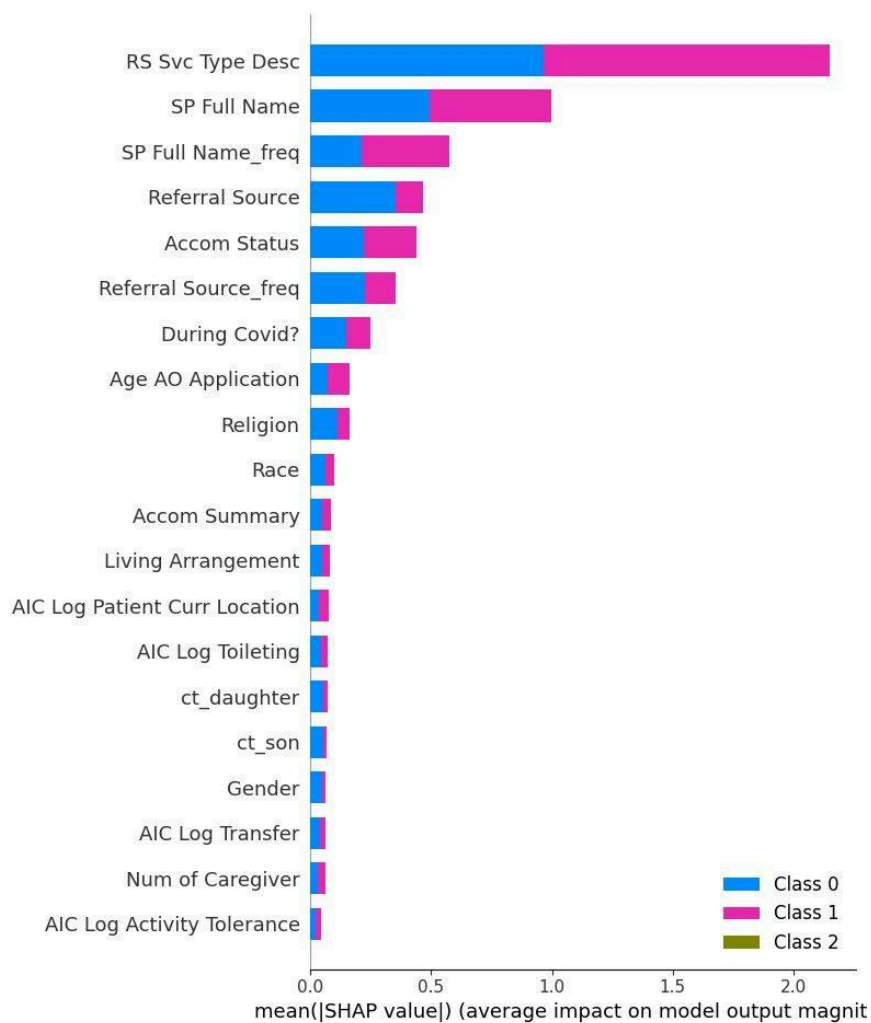


Figure A3-3: Mean absolute SHAP values by feature and class (Class 0 = Admit, Class 1 = Non-Admit, Class 2 = Other). RS Svc Type Desc dominates predictive importance across all classes.

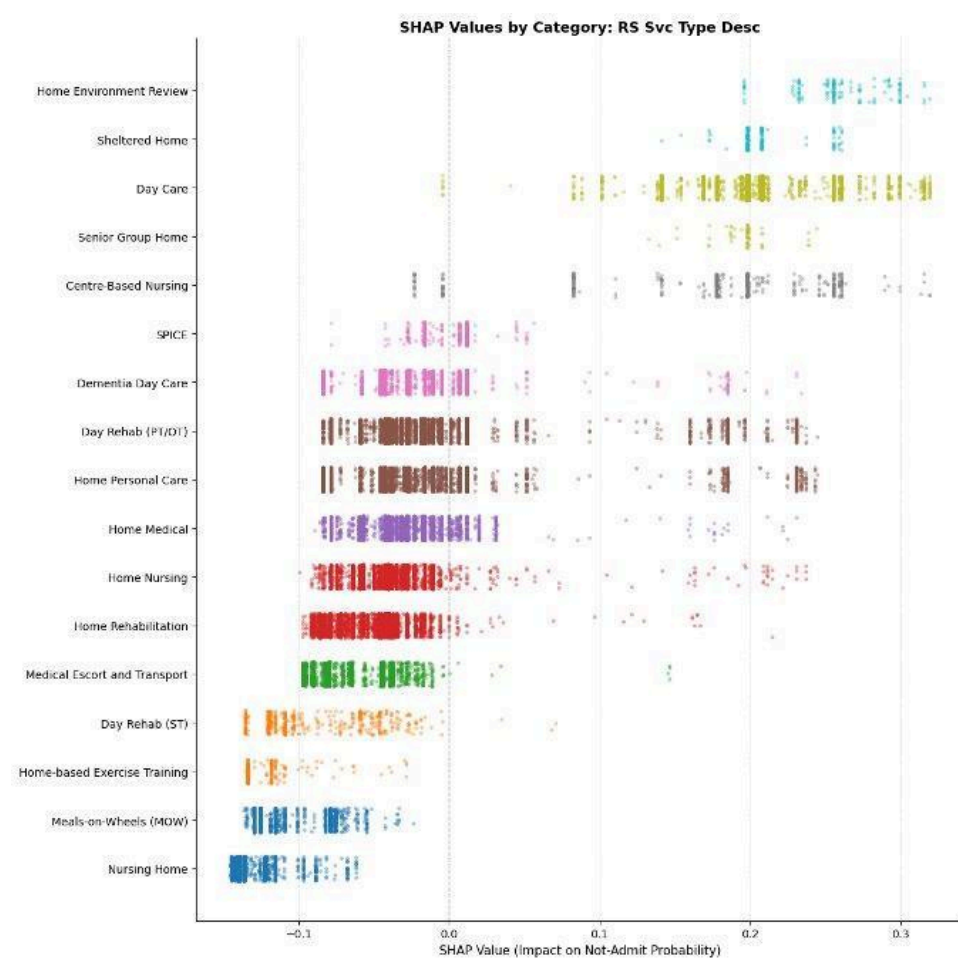


Figure A3-4: SHAP values disaggregated by service type (RS Svc Type Desc). Home Environment Review and Day Care are associated with Non-Admit outcomes; Nursing Home and Meals-on-Wheels are associated with Admit outcomes.

A4. Additional Exploratory Data Analysis

The following figures provide additional context on the patient population and dataset characteristics.

Figure 4: Patient Age Profile at Time of Referral

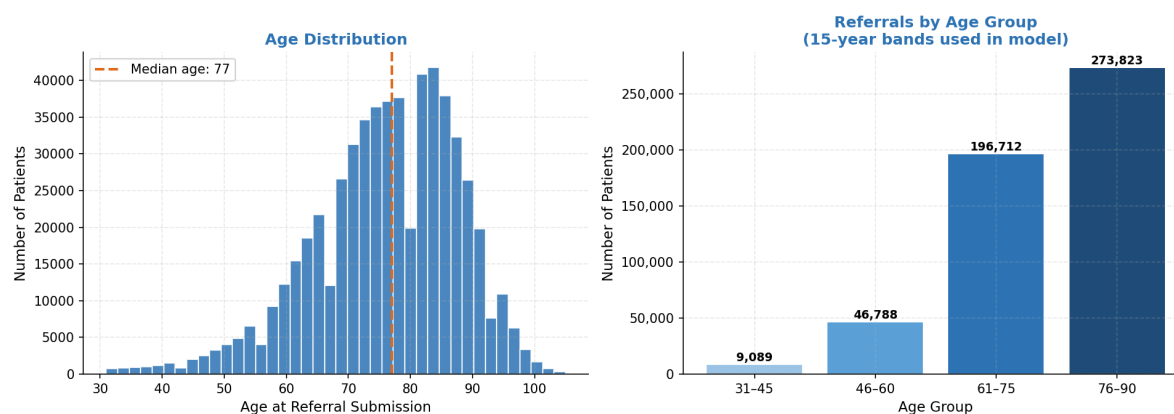


Figure A4-1: Patient age distribution at time of referral. The majority of referrals involve patients aged 61–90.

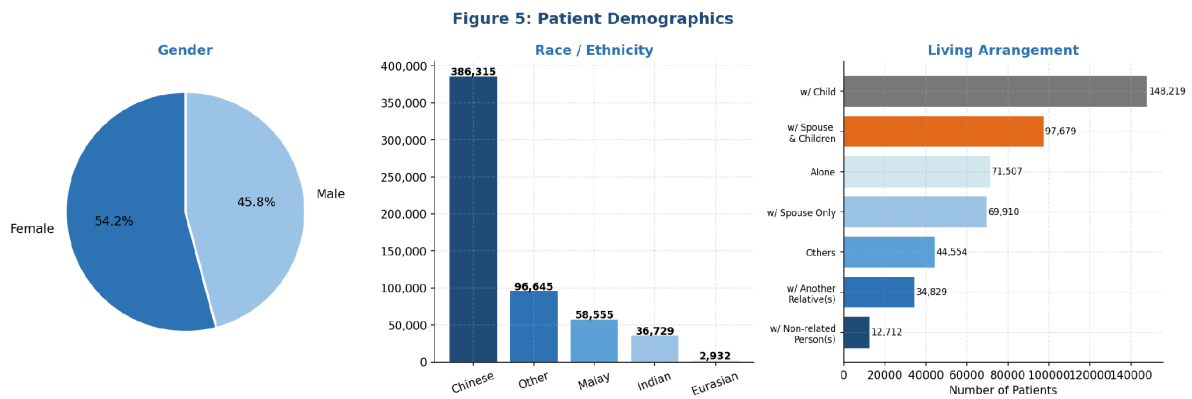


Figure A4-2: Patient demographic breakdown (race, citizenship, and related attributes).

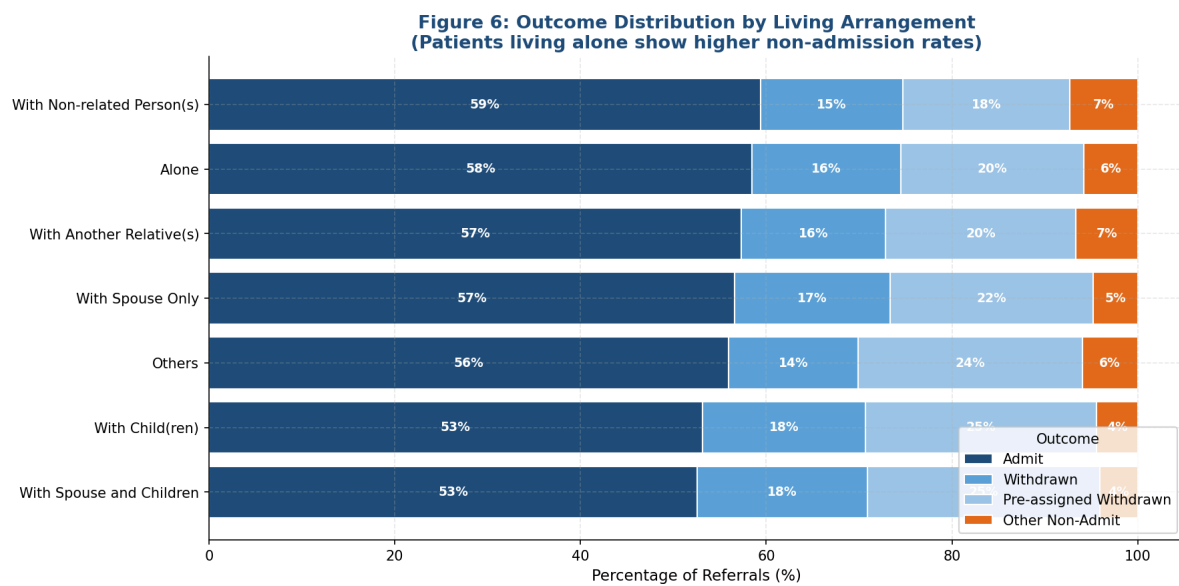


Figure A4-3: Referral outcome rates by living arrangement. Patients living alone show higher non-admission rates.

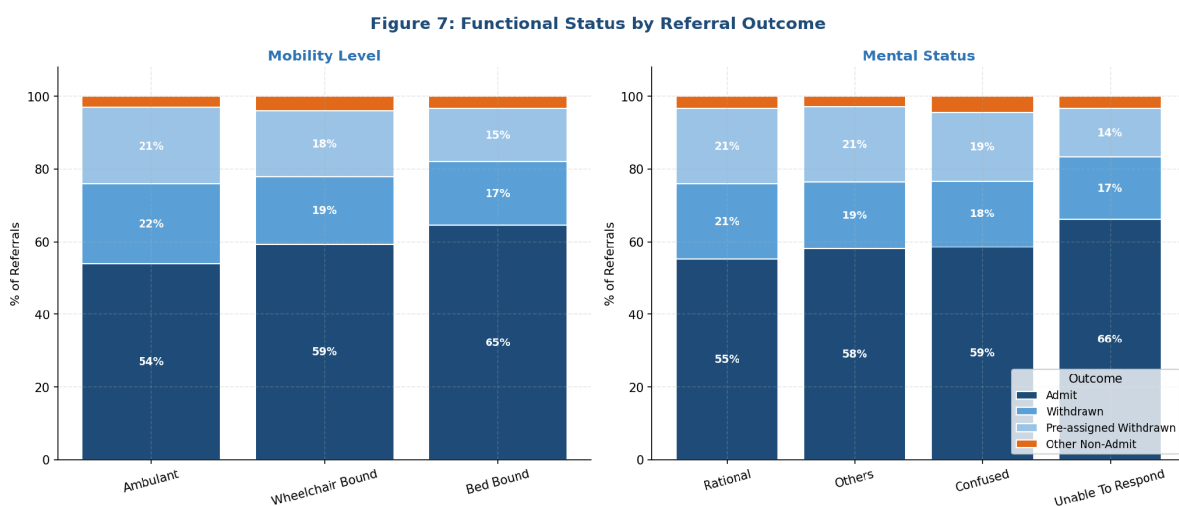


Figure A4-4: Distribution of functional status indicators across the patient cohort.

Figure 8: Caregiver Profile

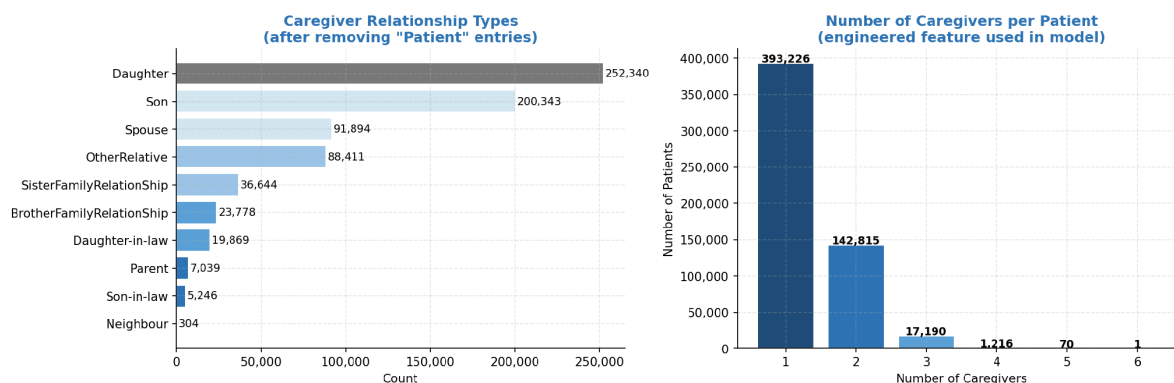


Figure A4-5: Caregiver type distribution and caregiver count per patient.

Figure 9: COVID-19 Impact on Referrals

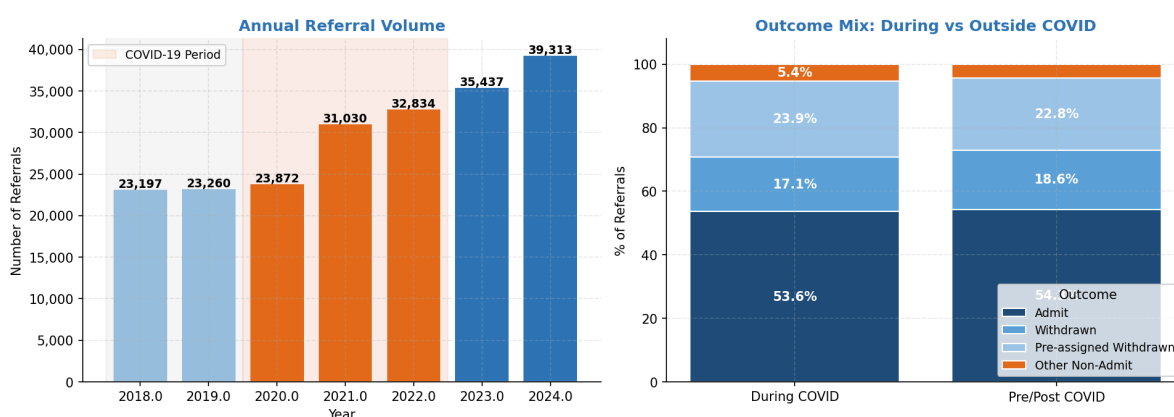


Figure A4-6: Referral outcome distributions before, during, and after the COVID-19 period.

A5. Full Flat Multi-Class Model Comparison

Results from Phase 1 flat multi-class classification across all five algorithms tested, referenced in Section 3.3.

Model	Accuracy	Balanced Acc.	Macro F1	Weighted F1
XGBoost	71.7%	26.2%	0.213	0.668
Random Forest	63.3%	38.2%	0.318	0.601
Logistic Regression	71.6%	14.3%	0.142	0.654
LightGBM	68.0%	31.9%	0.257	0.640
CatBoost	54.9%	37.9%	0.269	0.491

Table A5-1: Flat multi-class baseline performance across five algorithms.